

Lesson 21 — Build the Model & a Box That Holds a Number

Matches LEGO Teacher Portal lesson: "Space Scanner".

Objective (I can...): I can build the variables model, make a variable, and store a number in it. **CS concept:** Variables (a named box that holds a value) · **Builds on:** Lesson 20 (decisions we can now base on stored values).

AI Connection (2 min): AI stores what it learns as numbers/data. *First-principles question:* "What might a robot need to remember to get better at a task?"

Materials: LEGO CS & AI kit + device per pair, a labeled paper "box" per pair. **This is the unit's build lesson** — build the Variables model now and keep it assembled through Lesson 25.

60-minute flow - Warm-up (8): "Name the box" — coach holds a labeled box, drops in a number card, swaps it. A variable is a named box whose value can change. - **Concept + demo (10):** Create a variable "score," set it to 0, show it on screen, then change it and watch the display update. - **Build/Code (30):** Pairs **build the Variables model** (the unit's one build), then create a variable, set a starting number, and display it (screen or lights). *Swap at 15 min.* - **Challenge (7):** Set a different starting value; re-run to see the display change. - **Share + pack-up (5):** Pairs show their named variable and its stored value. **Leave the model built.**

Pair roles: Driver builds/codes, Navigator names the variable and reads its value. Swap at 15 min. (*Trio rotates.*)

Differentiation: *Support* — a twin example showing a named variable to adapt. *Extension / fast-finisher* — create two variables. *Reteach* — physically write a number in the paper box, then mirror it on screen.

Exit check: Pair creates a named variable and stores/displays a value in it. **At-home (optional):** Label 3 "boxes" (like "candles on cake"); give each a number. ~10 min.
